

A computer-certified 8/7 universal projective/injective gap

Run `fa_banach_001`, lane 15

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Status

This packet records a claimed partial result for the open problem of Aubrun, Lami, Palazuelos, Szarek, and Winter [1]: whether $\rho(X, Y) \geq \sqrt{2}$ for all real Banach spaces X, Y of dimension at least 2. The result below does not reach $\sqrt{2}$, but it improves the paper’s universal lower bound 19/18 to 8/7. This is exactly the computer-assisted improvement suggested, but not presented, in the remark after the proof of Theorem 2 of [1].

Problem 3. *Is it true that for any pair of Banach spaces X, Y with $\dim X, \dim Y \geq 2$, we have $\rho(X, Y) \geq \sqrt{2}$? In other words, does there always exist a nonzero tensor $z \in X \otimes Y$ such that*

$$\|z\|_{X \otimes_\pi Y} \geq \sqrt{2} \|z\|_{X \otimes_\varepsilon Y} ? \quad (31)$$

We now move on to the analysis of the asymptotic behaviour of the function $r_s(n)$. Developing functional analysis techniques from [48], we arrive at a lower estimate of $\rho(X, X)$ involving the Banach–Mazur distance of X from the Hilbert space of the same dimension (Theorem 16). Since

Figure 1: Problem 3 in [1], asking whether the universal gap is $\sqrt{2}$.

$$\|e_1 + e_2\|_X \leq \alpha, \quad \|e_1 - e_2\|_X \leq \beta$$

for some real numbers $\alpha, \beta \in [1, 2]$ such that $\alpha + \beta \leq 3$. This extra information can presumably be used to improve the lower bound in Theorem 2 to $\|z\|_{X \otimes_\pi Y} \geq \frac{8}{7} \|z\|_{X \otimes_\varepsilon Y}$ for an appropriate choice of z depending on α, β . However, since our arguments for that would rely heavily on computer assistance (and since the bound 8/7 is unlikely to be optimal), we do not present them.

Before we move on, let us discuss the special case of 2-dimensional spaces. Although we are not yet able to evaluate the two quantities $r(2, 2)$ and $r_s(2)$ exactly, we can show that

Figure 2: The source-paper remark suggesting the 8/7 improvement from the refined Auerbach data.

Definitions

For Banach spaces X, Y , let

$$\rho(X, Y) = \sup_{0 \neq z \in X \otimes Y} \frac{\|z\|_{X \otimes_\pi Y}}{\|z\|_{X \otimes_\varepsilon Y}}.$$

Equivalently, $\rho(X, Y)$ is the best domination constant of the injective norm by the projective norm on $X \otimes Y$. The source paper defines

$$r(n, m) = \inf_{\dim X = n, \dim Y = m} \rho(X, Y).$$

Proof Intuition

The proof of the 19/18 theorem in [1] uses a two-dimensional Auerbach argument and then tests a CHSH-type functional against a carefully chosen 2×2 tensor. The source paper deliberately discards half of the geometric information produced by the Auerbach step: besides one diagonal bound, the proof actually gives two bounds

$$\|e_1 + e_2\| \leq \alpha, \quad \|e_1 - e_2\| \leq \beta, \quad \alpha + \beta \leq 3.$$

Keeping both bounds changes the coordinate body from a hexagon with one strip constraint to a clipped square with two strip constraints. The tensor can then be chosen adaptively from a small linear program. The worst case occurs at the balanced octagon $\alpha = \beta = 3/2$, where the certificate is simply $(4/7)$ times the usual CHSH tensor. Away from the balanced case there is extra room, and the exact verifier checks the resulting finite family of polynomial inequalities.

Main Result

Theorem 1. *Let X and Y be real Banach spaces with $\dim X, \dim Y \geq 2$. Then*

$$\rho(X, Y) \geq \frac{8}{7}.$$

Consequently, $r(n, m) \geq 8/7$ for all $n, m \geq 2$.

Proof. We use the refined form of Lemma 15 from [1]. Applied to X , it gives vectors $e_1, e_2 \in X$ and functionals $e_1^*, e_2^* \in X^*$ such that

$$\|e_i\| = \|e_i^*\| = 1, \quad e_j^*(e_i) = \delta_{ij},$$

and, for some $\alpha, \beta \in [1, 2]$,

$$\|e_1 + e_2\| \leq \alpha, \quad \|e_1 - e_2\| \leq \beta, \quad \alpha + \beta \leq 3.$$

Apply the same construction in Y , obtaining f_1, f_2, f_1^*, f_2^* and parameters $\gamma, \delta \in [1, 2]$ with $\gamma + \delta \leq 3$.

If necessary, enlarge the parameters while keeping them in $[1, 2]$ until $\alpha + \beta = 3$ and $\gamma + \delta = 3$. This only enlarges the coordinate sets used below and therefore makes the injective-norm estimate harder. Thus it suffices to prove the boundary case

$$\alpha = 1 + m, \quad \beta = 2 - m, \quad \gamma = 1 + r, \quad \delta = 2 - r$$

with $m, r \in [0, 1]$. Define

$$H_m = \{(s, t) \in \mathbb{R}^2 : |s| \leq 1, |t| \leq 1, |s + t| \leq 1 + m, |s - t| \leq 2 - m\}.$$

For every $\phi \in B_{X^*}$, the coordinate pair $(\phi(e_1), \phi(e_2))$ lies in H_m , and similarly the Y -coordinates lie in H_r .

The following certified finite-dimensional lemma is the only computer-assisted part of the proof.

Lemma 1 (certified two-parameter CHSH certificate). *For every $m, r \in [0, 1]$ there exists a real 2×2 matrix*

$$A(m, r) = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

such that

$$\left| x^\top A(m, r) y \right| \leq 1 \quad \forall x \in H_m, \quad \forall y \in H_r,$$

and

$$a_{11} + a_{12} + a_{21} - a_{22} \geq \frac{16}{7}.$$

Assuming Lemma 1, set

$$z = \sum_{i,j=1}^2 a_{ij} e_i \otimes f_j \in X \otimes Y.$$

The first assertion of the lemma gives

$$\|z\|_{X \otimes_\varepsilon Y} \leq 1.$$

Now consider the CHSH functional

$$w^* = e_1^* \otimes f_1^* + e_1^* \otimes f_2^* + e_2^* \otimes f_1^* - e_2^* \otimes f_2^*.$$

Since e_i^* and f_j^* have norm 1, the usual CHSH inequality gives

$$\|w^*\|_{X^* \otimes_\varepsilon Y^*} \leq 2.$$

Therefore, by duality,

$$\|z\|_{X \otimes_\pi Y} \geq \frac{|w^*(z)|}{\|w^*\|_{X^* \otimes_\varepsilon Y^*}} \geq \frac{(16/7)}{2} = \frac{8}{7}.$$

Since $\|z\|_{X \otimes_\varepsilon Y} \leq 1$, this proves $\rho(X, Y) \geq 8/7$. □

Proof of the Certified Lemma

We give the exact finite certificate and point to the verifier implementing the arithmetic. On the boundary, the vertices of H_m are

$$\begin{aligned} &(1, m), (1, m-1), (-1, -m), (-1, 1-m), \\ &(m, 1), (m-1, 1), (-m, -1), (1-m, -1), \end{aligned}$$

with the analogous list for H_r . Let $A = A(m, r)$ be the unique solution of the four linear equations

$$\begin{aligned} (1, m)^\top A(1, r) &= 1, \\ (1, m-1)^\top A(r, 1) &= 1, \\ (m, 1)^\top A(1, r-1) &= 1, \\ (m-1, 1)^\top A(r-1, 1) &= -1. \end{aligned}$$

The entries of A are rational functions of m, r . The denominator is strictly positive on the square $[0, 1]^2$. Because H_m and H_r are polytopes and the form $x^\top A y$ is affine in each variable, it is enough to check the $8 \cdot 8$ vertex pairs. Thus Lemma 1 reduces to the following exact inequalities on $[0, 1]^2$:

$$\begin{aligned} D(m, r) &> 0, \\ (1 \pm v_i(m)^\top A(m, r) v_j(r)) D(m, r) &\geq 0 \quad (0 \leq i, j < 8), \end{aligned}$$

and

$$\left(a_{11} + a_{12} + a_{21} - a_{22} - \frac{16}{7}\right) D(m, r) \geq 0.$$

The verifier

`code/verify_8over7_certificate.py`

symbolically constructs these 130 polynomials and proves their nonnegativity using exact rational Bernstein coefficients with recursive subdivision. The run included in this packet reports:

`checks=130, failures=0, max_depth_used=2.`

For orientation, at the balanced point $m = r = 1/2$ the certificate is

$$A = \frac{4}{7} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

and the inequality is tight for this certificate:

$$a_{11} + a_{12} + a_{21} - a_{22} = \frac{16}{7}.$$

Analytic Audit of the Certificate

The helper

`code/analyze_certificate_symmetry.py`

gives a more reviewable view of the same exact certificate. Under the symmetries $m \leftrightarrow 1 - m$, $r \leftrightarrow 1 - r$, and $m \leftrightarrow r$, the 130 polynomial checks collapse to 10 representative classes. Each representative is then checked on the four quadrants of the centered square by exact Bernstein coefficients.

The most important representative, the objective inequality, has a simple hand proof. Put $x = 2m - 1$, $y = 2r - 1$, $X = x^2$, and $Y = y^2$. After clearing the positive scalar factor, the numerator of the objective class is

$$X^2Y^2 - 10X^2Y + 9X^2 - 10XY^2 + 51XY + 15X + 9Y^2 + 15Y.$$

On $0 \leq X, Y \leq 1$ this equals

$$X^2(1 - Y)(9 - Y) + X(-10Y^2 + 51Y + 15) + 9Y^2 + 15Y,$$

which is nonnegative. Thus the tight part of the certificate is already elementary; the remaining nontrivial vertex classes are the part still best audited through the exact Bernstein atlas.

Verification Notes

The proof uses only:

- Lemma 15 and its stated refinement from [1];
- the elementary CHSH bound for four scalar variables in $[-1, 1]$;
- the exact rational verifier included with this packet.

The verifier does not call a numerical optimizer. The earlier LP searches were used only to discover the four defining equations for $A(m, r)$.

Limitations and Novelty Check

This does not solve Problem 3 of [1]: the conjectural sharp lower bound $\sqrt{2}$ remains open. The result is a universal constant improvement

$$19/18 \longrightarrow 8/7.$$

Local run indexes were checked for arXiv id 1809.10616, the title, “projective/injective ratio”, “XOR games”, “sqrt(2)”, and “8/7”; no existing promoted packet was found. A bounded web search for exact phrases around $\rho(X, Y)$, “projective-injective ratio”, “sqrt(2)”, and “8/7” found the source paper but no obvious later paper presenting this certificate or resolving the $\sqrt{2}$ problem. Novelty confidence is therefore moderate-high, subject to expert literature review.

Human Review Recommendation

Recommended review focus:

1. check that the refined Auerbach data from Lemma 15 of [1] indeed applies independently to X and Y in the form used here;
2. audit the exact verifier, especially the four equations defining $A(m, r)$, the vertex list for H_m , and the Bernstein nonnegativity routine;
3. confirm that enlarging to the boundary $\alpha + \beta = \gamma + \delta = 3$ is legitimate for the injective-norm upper bound.

If these checks pass, this packet should be treated as a likely valid partial result toward Problem 3.

References

- [1] G. Aubrun, L. Lami, C. Palazuelos, S. J. Szarek, and A. Winter, *Universal gaps for XOR games from estimates on tensor norm ratios*, arXiv:1809.10616.